LLM-Based Synthetic Tabular Data Generation and Curation for Privacy-Sensitive Scientific Applications

***Abstract*—Scientific discovery increasingly depends on high-quality, AI-ready datasets, yet data scarcity remains in a major fields of science. This work presents a two-stage approach combining LLM-based generation with curation that addresses data scarcity in a privacy-sensitive domains as medicine and social sciences. Through extensive evaluation across multiple model scales, we demonstrate that large-scale LLMs consistently outperform specialized generative models in statistical fidelity and downstream task utility, but introduce substantial privacy vulnerabilities. Our automated curation pipeline effectively balances privacy-utility trade-offs, reducing privacy risks while improving data quality. However, synthetic data augmentation reveals quality degradation for most predictive models, with notable exceptions: fine-tuned LLaMA demonstrates stable classification performance enabling moderate data augmentation, while TabPFN emerges as uniquely robust to synthetic artifacts across all conditions. LLM fine-tuning on synthetically augmented data shows dramatic privacy improvements but substantial performance challenges, establishing the methodology as primarily valuable for privacy-sensitive scientific applications where performance trade-offs are acceptable.**

***Index Terms*—synthetic data, large language models, data curation, data augmentation, AI-ready datasets, tabular data, machine learning**

I. Introduction

Synthetic tabular data generation represents a fundamental challenge in machine learning, with applications spanning privacy-preserving data sharing, data augmentation, and statistical modeling. Contemporary approaches rely predominantly on specialized generative models, including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models. However, these methods exhibit critical limitations when modeling complex multidimensional distributions with intricate dependencies.

Scientific discovery is entering a new era powered by artificial intelligence and machine learning, yet one major bottleneck remains: the lack of high-quality, domain-specific datasets that are truly AI-ready. In physics, chemistry, biology, and materials science, researchers face critical data scarcity that limits the application of modern ML techniques to accelerate discovery processes. Traditional experimental data collection is expensive, time-consuming, and often limited by ethical or safety constraints.

Recent advances in large language models (LLMs) have demonstrated remarkable capabilities across diverse domains, yet their potential for creating AI-ready scientific datasets tabular data synthesis remains largely unexplored in scientific contexts. Previous investigations have been constrained to demonstrating LLM generation quality of tabular data without addressing the critical gap between synthetic data creation and practical machine learning utility models with focus on small LLM (with fewer than 1 billion parameters - e.g. GREAT [1], PAFT [2]), that limiting our understanding of scaling and emergent capabilities of LLM in tabular synthesis domain.

The creation of AI-ready scientific datasets requires not only synthetic data generation but also intelligent curation processes that ensure statistical consistency, eliminate artifacts, and optimize data utility for downstream applications. Unlike raw synthetic data, AI-ready datasets must undergo systematic quality assessment, refinement, and validation to ensure reliable performance in scientific machine learning applications.

Our primary contributions include: (1) a novel automated curation pipeline that systematically balances privacy-utility trade-offs through multi-objective optimization, improving data quality while reducing privacy risks, (2) comprehensive evaluation revealing that large-scale LLMs outperform specialized models in generation quality but create inherent privacy vulnerabilities that curation only partially mitigates, (3) systematic assessment demonstrating expected quality degradation in synthetic data augmentation with promising exceptions—fine-tuned LLaMA showing superior stability for classification tasks and TabPFN demonstrating exceptional robustness to synthetic artifacts compared to traditional architectures, (4) evidence that LLM fine-tuning on synthetic data enables substantial privacy preservation while causing significant performance challenges, establishing this approach as primarily suitable for privacy-sensitive scientific domains rather than general-purpose enhancement of machine learning capabilities.

II. Related Work

A. Synthetic Tabular Data Generation

Recent years have witnessed significant developments in synthetic tabular data generation methodologies. Bayesian methods employ probabilistic models that ensure high interpretability but face substantial limitations when dealing with high-dimensional data or complex nonlinear dependencies. Generative Adversarial Networks (GANs) have introduced specialized architectures including CTGAN [3], PATEGAN [4], ADSGAN [5], and GANBLR [6] that address various aspects of tabular generation challenges. However, these GAN-based approaches frequently encounter training instability and mode collapse issues, limiting their reliability in production environments. To mitigate some of these issues, models like VECTGAN [7] have been proposed, which leverage vectorized embeddings to better capture complex feature interactions and improve training stability in tabular data synthesis.

Variational Autoencoders (VAEs) represent an alternative class of generative models with specialized versions for tabular data, including TVAE [8] and RTVAE [9]. Recent diffusion models, particularly TabDDPM [10], demonstrate impressive capabilities in modeling complex distributions through iterative denoising processes but require substantial computational resources.

The emergence of Large Language Models (LLMs) has introduced novel possibilities for tabular data synthesis. GREAT [1] pioneered this direction by adapting language models for tabular data through textual encoding. PAFT [2] advanced the field by introducing functional dependency awareness, while recent work [11] demonstrated LLM superiority over specialized models in preserving statistical properties. Additionally, GOGGLE [12] represents a novel approach that combines graph-based modeling with transformer architectures, enabling effective learning of relational dependencies in tabular data and enhancing generation quality.

B. Data Curation for AI-Ready Datasets

Data curation methodologies have emerged as critical components for creating reliable datasets for machine learning applications. Traditional approaches focus on manual quality assessment and expert-driven validation processes. However, the scale and complexity of modern scientific datasets necessitate automated approaches that can systematically identify and correct data quality issues. Data curation is a multifaceted process aimed at improving the overall quality and reliability of datasets. It can encompass data cleaning (filtering), imputation of missing values, and correction or refinement of labels (label enhancement). During the filtering stage, the primary objective is to eliminate outliers and anomalous values that exert a detrimental impact on data quality. The curation of synthetic tabular data introduces unique challenges. Unlike in the domain of synthetic data generation for question answering, where the validity of a sample can often be explicitly verified, tabular settings lack a definitive ground-truth criterion for determining whether a given instance is of high quality. Consequently, the evaluation of synthetic tabular data is inherently more complex. In practice, various heuristic metrics are employed for this purpose. For instance, model-driven evaluation can serve as a practical proxy for data curation quality, with popular choices including gradient-boosted decision trees (e.g., XGBoost) and deep neural networks (Seedat et al., 2023) [13]. Additionally, large language models (LLMs) have recently been explored as filtering mechanisms (Chen et al., 2023) [14], given their strong contextual reasoning abilities, which enable them to effectively discard low-quality samples. A further complication arises from the generation process itself. In particular, LLMs are prone to hallucinations, which may result in the creation of non-existent class labels or unrealistic values that do not align with the inherent data distribution. This motivates the need for systematic label enhancement. One promising strategy for mitigating annotation costs involves replacing manual labeling with auxiliary student models that support knowledge distillation and iterative label refinement (Saad-Falcon et al., 2023) [15]. The underlying assumption is that student models distilled from LLM teachers can outperform conventional weak supervision by producing more robust and reliable annotations. As LLMs continue to advance in capability and accessibility, the integration of such student models is poised to emerge as a cost-effective and scalable alternative to human annotation.

C. Machine Learning with Synthetic Data

The utility of synthetic data for machine learning applications depends critically on both generation quality and post-processing methodologies. Recent studies have demonstrated that raw synthetic data often requires additional refinement to achieve optimal performance in downstream tasks. Data augmentation strategies using synthetic data have shown promise in addressing data scarcity challenges, particularly in scientific domains where experimental data collection is expensive or limited.

Transfer learning with synthetic data presents unique opportunities for scientific applications, where models trained on synthetic datasets can potentially generalize to real-world scenarios. However, the effectiveness of this approach depends heavily on the quality and representativeness of the synthetic training data, highlighting the importance of intelligent curation processes.

Recent advancements in generating synthetic tabular data have highlighted the relevance of scalable and privacy-concealing approaches safeguarding utility in the data while concealing sensitive information. Liu et al. [16], for example, introduce end-to-end synthetic data generation frameworks specifically designed for learning analytics such that a proper balance between scalability as well as preservation of privacy is maintained for efficient use in data analytics as well as data mining applications. These results indicate that leading-edge synthetic data generators should incorporate proper evaluation schemes in order to sustain utility in the data for machine learning applications.

Further, foundation models for tabular data have emerged as extremely effective in making precise prediction despite having few available initial datasets. Hollmann et al. delve into a foundation model for tabular data called TabPFN [17] founded on synthetic data construction methods for coping with few available datasets while enhancing model generalizability. Their research highlights the focal importance of high-quality synthetic datasets for efficient performance for foundation models over downstream scientific tasks with an emphasis on a necessary careful curation and fine-tuning of synthetic datasets before model training.

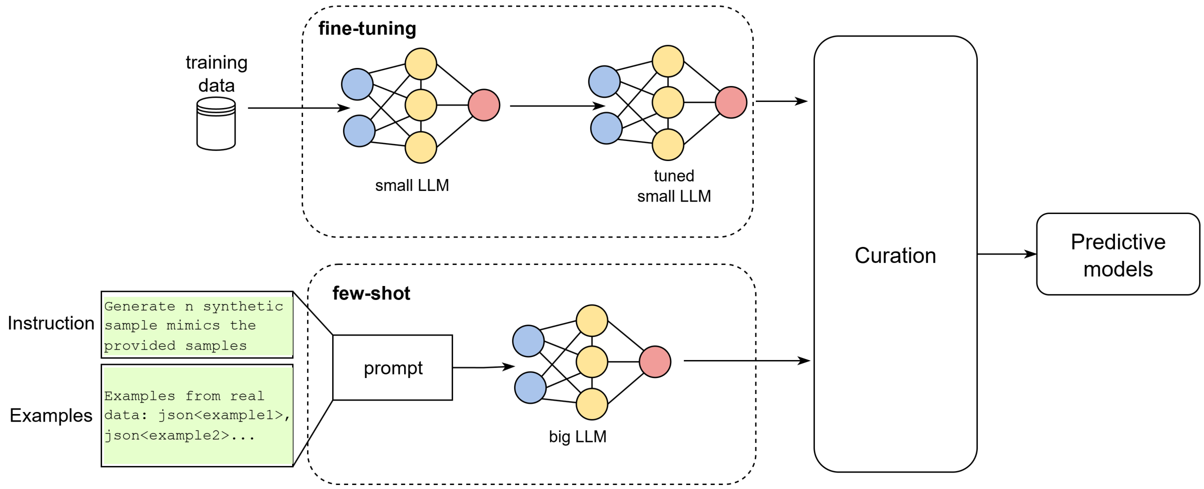

Figure 1. Overview of the complete pipeline: synthetic data generation methods (few-shot LLM-based generation, fine-tuning of small LLMs), data curation, and training of predictive models.

III. Methodology

A. Enhanced LLM-based Synthetic Data Generation

We formalize tabular data synthesis as the problem of generating synthetic dataset $\mathcal{D}_{syn} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ that preserves the multidimensional joint distribution $P(\mathbf{X}, Y)$ of the original dataset $\mathcal{D}_{real}$, where $\mathbf{X} \in \mathbb{R}^d$ represents feature vectors and Y represents target variables.

Based on observed scaling laws in language models, we implement differentiated approaches according to model size.

For GREAT, PAFT we implement intensive fine-tuning with causality-aware adaptations using LoRA [18] for parameter-efficient training. Each batch contains of ten example rows per training prompt to enhance pattern learning from multiple instances and improve statistical relationship modeling.

For Llama-3.2-3B we implement LoRA fine-tuning with batch strategy (Llama BT): Ten example rows per training prompt to enhance pattern learning from multiple instances and improve statistical relationship modeling.

Our fine-tuning strategy adapts PAFT (Permutation-aided Fine-Tuning) with explicit causal structure consideration, implementing target-conditioned ordering that maintains functional dependencies. Two models were chosen for fine-tuning: DistilGPT-2 82M and Llama-3.2-3B [19]. For computational efficiency, we employ LoRA with rank $r = 16$ and scaling factor $\alpha = 32$ for DistilGPT-2 and $\alpha = 16$ for Llama-3.2-3B to enable parameter-efficient training while preserving pre-trained knowledge.

For models exceeding 30B parameters (Qwen-3 32B, LLaMA-3.1 70B, GPT-4o), we leverage emergent in-context learning capabilities without additional training. Data representation employs the "column_name is value" format for consistency with language modeling objectives. The prompt structure comprises: (1) direct generation instructions specifying row-by-row synthesis, (2) target row count specification, and (3) representative data exemplars in JSON format for enhanced structural clarity. Data preprocessing includes missing value removal, label encoding for categorical features, and four-decimal precision rounding for numerical values. Generation employs batch processing with 10 examples per batch to balance context utilization and computational efficiency, preventing context window overflow while maintaining sufficient pattern exposure.

The LLM evaluation spectrum spans five orders of magnitude: DistilGPT-2 82M (GReaT, PAFT), Llama-3.2-3B BT, LLaMA-70B, Qwen 32B and GPT-4o. This coverage enables comprehensive scaling analysis from sub-billion parameter models requiring intensive fine-tuning to large-scale models demonstrating strong few-shot capabilities.

We evaluate LLM tabular generations against 9 specialized baseline models representing current state-of-the-art in tabular generation: Bayesian Networks, GAN variants (CTGAN, PATEGAN, ADSGAN, GANBLR++), VAE approaches (TVAE, RTVAE), and diffusion models (TabDDPM). All specialized models were tuned with Optuna hyper-parameter optimization.

B. Intelligent Data Curation Pipeline

Our curation pipeline (Figure 1) addresses the critical gap between raw synthetic generation and AI-ready data utility through systematic automated quality enhancement.

The quality assessment module implements multi-dimensional evaluation of synthetic data characteristics. Data curation is performed on the basis of synthetic data quality metrics. In other words, the curation problem can be reduced to a task of multi-objective optimization with respect to these metrics. To facilitate the search for an optimal solution, it is necessary to reduce the number of metrics employed in the optimization process; dimensionality reduction is also required to guarantee the existence of at least one feasible optimal solution.

As primary metrics, we select the ROC AUC score for a linear model f_L and for a gradient boosting model f_{GB} (and the coefficient of determination R^2 in the regression setting).

This choice is motivated by the fact that the applicability of synthetic tabular data for predictive modeling remains one of the central questions under investigation in contemporary machine learning research. To reduce the dimensionality of the optimization space, we represent the metrics as a weighted sum f_{ML}:

$$f_{ML}(x) = w_1 f_L(x) + w_2 f_{GB}(x)$$

where w_1 and w_2 are the coefficients of a convex combination of the metrics. In the general case, $w_1 \neq w_2$, which makes it possible to assign higher priority to either the linear model or gradient boosting, depending on the task at hand. For simplicity, however, we assume equal weights for both metrics.

Optimization exclusively over machine learning performance metrics may lead to degradation of other important criteria, particularly privacy, which is unacceptable when working with sensitive datasets in domains such as healthcare or finance. Therefore, we introduce a privacy metric f_{pr} and formulate the curation task as a multi-objective optimization problem between the aggregated machine learning metric f_{ML} and the privacy metric f_{pr}. The subsample indices ξ are drawn randomly from a uniform distribution:

$$\max_{X^* \subset X} \{f(x, \xi) = (f_{ML}(x, \xi), f_{pr}(x, \xi))\}$$

where X^* is a set of optimal solutions. To reduce the computational cost of searching for an optimal solution, subsample indices are generated using the Monte Carlo method. The optimal solution is then determined according to the Pareto optimality criterion, which allows one to identify a compromise between machine learning performance and privacy without substantial degradation of either metric.

C. ML Training on synthetic data

1) Augmented Training Strategies: Our evaluation framework encompasses three primary training scenarios: training on original real data to establish baseline performance, training on mixture of synthetic and real data to assess basic generation utility, and training on curated synthetic data to evaluate curation effectiveness.

Data combination methodologies explore optimal mixing ratios between real and synthetic data.

2) Prediction models Training: Model selection includes XGBoost as traditional machine learning approach, MLP and TabPFN as SOTA deep learning architecture for tabular predictions.

3) LLM Few-shot and Fine-tuning approaches: The proportion of augmented synthetic data, as well as the application of curation, influences the selection of in-context examples utilized by the few-shot approach. Furthermore, the Llama-3.2-3B model will be fine-tuned anew on each data variant, enabling an assessment of the effectiveness of the generated synthetic data for applications in scientific domains.

IV. Experimental Setup

A. Datasets and Experimental Design

Our experimental framework encompasses 6 datasets covering both classification (4 datasets) and regression (2 datasets) tasks (Table I), with a focus on capturing diverse scientific patterns. All datasets are of scientific nature, comprising 3 from the medical domain and 3 from the sociological domain, ensuring a balanced representation across disciplines.

Dataset characteristics range from small-scale (hundreds of samples) to medium-scale (thousands of samples) experimental datasets across varying dimensionalities and feature types. Each dataset undergoes systematic preprocessing including missing value handling, feature scaling, and quality assessment to establish baseline characteristics.

The only exception is the initial set of experiments aimed at demonstrating and comparing the capabilities of different models in generating synthetic tabular data. For these experiments, an extended set of datasets was employed: in addition to the aforementioned datasets, the evaluation was also conducted on the following: *iris, wine, 560_bodyfat, travel, credit_g, phoneme, page_blocks, 562_cpu_small, pendigits, magic04, online_shoppers, heloc_dataset_v1*. In addition to publicly available datasets, there were also close/rare domain-specific datasets that were collected by authors with industrial partners. They include:

- **social_net_interest** and **social_net_age** represent a single dataset with both classification and regression targets, containing anonymized and open data from users of a social network, enriched with additional information from social survey results.
- **oil_netpay** and **oil_lithology** comprise a single dataset with both classification and regression targets, consisting of data openly released by several oil companies.
- **flood** is a closed dataset containing spatial distribution of sea level data in a bay area.

Table I
Dataset characteristics for synthetic data generation

Dataset Name	Rows	Columns	Elements	Task Type
Sociological datasets				
seattle_housing	2521	8	20168	Reg.
us_location	20400	5	102000	Class.
adult	48842	15	732630	Class.
Medical datasets				
diabetes	442	11	4862	Reg.
breast_cancer	569	31	17639	Class.
nursery	12958	9	116622	Class.

B. Compared Synthetic Generation Models and Evaluation Metrics

We evaluate against specialized baseline models representing current state-of-the-art in tabular generation: Bayesian Networks, GAN variants (CTGAN, PATEGAN, ADSGAN), VAE approaches (TVAE, RTVAE), and diffusion models (TabDDPM). Additionally, an assessment was performed on advanced generative models GANBLR and GANBLR++. However, their performance was significantly inferior compared to other generative approaches in terms of statistical fidelity and downstream task utility. Although GANBLR and GANBLR++ demonstrated relatively higher privacy scores, this advantage did not offset their lower effectiveness in overall utility and data quality metrics. Consequently, due to these comparatively poor results—reflected by their consistently lower rankings across multiple critical evaluation criteria—these models were excluded from the overall comparative review. The LLM evaluation spectrum includes DistilGPT-2 82M, Llama-3.2-3B, LLaMA-70B, Qwen 32B and GPT-4o.

Evaluation employs Synthcity [20] metrics suite across five categories: *sanity* (data validity checks), *statistical* (distributional fidelity measures), *performance* (downstream task utility), *detection* (synthetic data detectability), and *privacy* (privacy preservation assessment).

V. Results and Analysis

A. LLM Privacy-Utility Trade-off for Synthetic Data Generation

Our evaluation confirms the superiority of large-scale LLMs over specialized generative models across several quality metrics (in Table II). Few-shot LLMs achieve superior statistical fidelity with GPT-4o demonstrating PRDC precision of 1.0 and recall of 1.0, compared to 0.4-0.8 precision and 0.4-1.0 recall for specialized models. Large LLMs also show enhanced distributional coverage with PRDC Coverage scores of 0.9 (LLAMA 70B, GPT-4o, Qwen 32B) compared to 0.1-0.8 for traditional approaches.

The performance analysis shows systematic patterns where few-shot large LLMs consistently achieve top-tier rankings in downstream task utility. LLMs demonstrate superior XGB Syn LLA performance with scores of 0.9 (LLAMA 70B, GPT-4o, Qwen 32B) compared to 0.4-0.8 for specialized models. Model scale appears more important than task-specific fine-tuning, with few-shot large models avoiding instabilities that affect smaller fine-tuned variants (llama ST: 0.7, PAFT: 0.3-0.5).

The detection metrics indicate comparable detectability across most approaches, with GPT-4o achieving MLP detection scores of 0.0, similar to most specialized models (0.0-0.1), though DPGAN shows higher detectability at 0.3.

Although this superior performance comes with significant privacy costs. The privacy evaluation reveals systematic vulnerabilities across large language models compared to specialized generative approaches (Table III). While specialized privacy preserving model DPGAN maintains low identifiability risk, GPT-4o exhibits an identifiability score of 0.5, much worser than traditional models (0.1-0.3). This elevated identifiability indicates that LLM-generated samples are substantially more traceable to original training data, suggesting increased memorization of input patterns.

The following analysis demonstrates how automated curation can partially bridge this privacy-utility gap across different generative approaches.

B. Data Curation Impact Assessment

1) Quality Improvement Analysis: Data curation demonstrates significant quality improvements across multiple metrics.

Quality score distributions show systematic improvement post-curation (Table IV). We achieved an average reduction of approximately 20 % in identifiability across all evaluated models, thereby decreasing the likelihood of confidential information leakage to potential adversaries. Regarding classification metrics, most models demonstrated improvements in ROC AUC, with consistent gains of up to 0.26 points. For large-scale models such as LLaMA 70B and Qwen 32B, the performance improvements were marginal; however, we succeeded in reducing privacy-related risk without degrading performance. For the remaining models, we identified configurations that simultaneously improved both performance and privacy.

A similar trend was observed for regression tasks: after curation, the coefficient of determination became positive for the majority of models, while the Mean Absolute Percentage Error (MAPE) decreased significantly. The most substantial improvement was observed for DDPM, with MAPE reduced by 40 %. These results indicate that the synthetic dataset contains high-quality samples suitable for predictive modeling tasks, provided that appropriate preprocessing is applied.

The changes in statistical metrics following curation were negligible, with an average deviation of no more than 0.02 compared to non-curated data. Notably, we achieved a substantial reduction in the Wasserstein distance for DDPM-generated samples. This suggests that improvements in both performance and privacy were obtained without explicit degradation of the underlying data distribution, despite the absence of a direct constraint enforcing distributional preservation.

C. Predictive Model Performance on Synthetic Data

1) Classification Task Performance: Figure 2 shows the results of assessing the quality of classification models (MLP, TabPFN, XGBoost) when trained on data supplemented with synthetics, with different ratios of real and synthetic data (70/30, 50/50, 30/70). The results are shown for both synthetics before curation (not) and after curation (cur). The last column corresponds to the base models trained exclusively on real data (train_real). The quality metric is ROC AUC.

The three models (MLP, TabPFN, XGBoost) achieve high performance on classification datasets with mean ROC AUC values greater than 0.8 across most conditions. The TabPFN model shows the highest stability because its ROC AUC scores remain between 0.95 and 0.98 regardless of synthetic data inclusion. The ROC AUC scores of MLP and XGBoost models decrease when synthetic data proportions increase from 30% to 70%, especially in uncurated data scenarios. The addition

Table II
COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: PERFORMANCE AND DETECTION METRICS. **GREEN BOLD** VALUES INDICATE BEST PERFORMANCE, **RED BOLD** VALUES INDICATE WORST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Evaluation Metric	GAN Models				VAE Models		Prob Graph		Diffusion	Large LLM			Fine-tuned LLM $\leq$ 3B		
	ctgan	adsgan	vectgan	dpgan	tvae	rtvae	bn	goggle	ddpm	LLAMA 70B	GPT-4o	Qwen 32B	llama BT	GReaT	PAFT
STATS METRICS															
Inv KL Divergence ↑	0.94	0.87	0.92	0.35	0.89	0.64	0.91	0.42	0.88	0.93	**0.98**	0.86	0.78	0.34	0.87
PRDC Coverage ↑	0.72	0.68	0.64	**0.14**	0.71	0.52	0.63	0.32	0.78	**0.89**	**0.88**	**0.91**	**0.92**	0.73	0.82
PRDC Density ↑	0.64	0.57	0.53	**0.18**	0.62	0.78	0.58	0.67	0.72	0.88	**0.98**	**0.97**	**1.05**	0.74	0.71
PRDC Precision ↑	0.78	0.73	0.64	0.42	0.79	0.75	0.68	0.72	0.81	0.87	**1.00**	**1.00**	**1.00**	0.73	0.69
PRDC Recall ↑	**1.00**	0.87	**1.00**	0.52	**1.00**	0.64	0.88	0.43	0.92	0.91	**1.00**	0.94	0.93	**1.00**	**1.00**
Wasserstein Distance ↓	0.14	0.12	0.13	**1.05**	0.11	0.09	0.17	0.08	0.87	0.23	0.12	0.19	0.24	0.58	0.32
PERFORMANCE METRICS															
Rank Dist Corr ↑	0.18	0.14	0.12	**0.04**	0.23	0.17	0.26	**0.03**	0.34	0.21	0.32	0.28	0.36	0.24	0.19
MLP Syn LLA ↑	0.57	**0.62**	0.58	0.43	0.64	0.63	0.61	0.38	0.59	0.63	0.64	0.57	0.42	0.48	0.34
XGB Syn LLA ↑	0.78	0.82	0.77	0.42	0.83	0.79	0.81	0.47	0.84	0.87	0.88	0.86	0.83	0.79	0.53
XGB Syn OOD ↑	0.82	0.87	0.83	0.41	0.88	0.84	0.86	0.52	0.83	0.92	0.94	0.91	0.85	0.87	0.51
DETECTION METRICS															
Detection MLP Mean ↓	**0.04**	0.12	0.09	0.27	**0.03**	**0.02**	0.11	0.18	**0.01**	0.08	**0.02**	**0.03**	**0.04**	**0.03**	**0.02**
Detection XGB Mean ↓	0.38	0.42	0.36	0.47	0.43	0.41	0.34	0.48	0.24	0.32	0.23	0.21	0.19	0.22	0.17

Table III
COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: SANITY AND PRIVACY ASSESSMENT. **GREEN BOLD** VALUES INDICATE BEST PERFORMANCE, **RED BOLD** VALUES INDICATE WORST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Evaluation Metric	GAN Models				VAE Models		Prob Graph		Diffusion	Large LLM			Fine-tuned LLM $\leq$ 3B		
	ctgan	adsgan	vectgan	dpgan	tvae	rtvae	bn	goggle	ddpm	LLAMA 70B	GPT-4o	Qwen 32B	llama BT	GReaT	PAFT
SANITY AND PRIVACY METRICS															
Common Rows Prob ↓	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**	**0.00**	**0.00**
Data Mismatch ↓	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	0.24	**0.38**	**0.42**
Distant Values Prob ↓	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Nearest Neighbor Dist ↓	0.23	0.18	0.21	**0.32**	0.19	0.22	0.24	0.17	0.21	0.16	0.14	0.18	0.22	**0.12**	**0.13**
Identifiability ↓	0.34	0.28	0.31	**0.14**	0.32	0.27	0.29	0.23	0.38	0.42	**0.47**	**0.48**	**0.46**	0.43	0.41

Table IV
COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS BEFORE AND AFTER CURATION. **GREEN BOLD** VALUES INDICATE BEST PERFORMANCE, **RED BOLD** VALUES INDICATE WORST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Evaluation Metric	ctgan		ddpm		GPT-4o		LLAMA 70B		Qwen 32B		llama BT	
	no cur	cur	no cur	cur	no cur	cur	no cur	cur	no cur	cur	no cur	cur
Close Values Prob ↑	0.69	0.68	0.54	**0.60**	0.70	0.71	0.69	0.70	0.68	0.73	0.68	0.70
Wasserstein Distance ↓	0.25	0.27	4.79	**3.60**	0.19	0.31	0.32	0.31	0.19	0.32	0.29	0.23
Inv KL Divergence ↑	0.75	0.69	0.67	0.62	0.80	0.79	0.77	0.75	0.79	0.77	0.66	0.63
KS test ↑	0.87	0.83	0.78	0.73	0.90	0.88	0.85	0.85	0.88	0.87	0.85	0.81
JS Divergence ↓	0.02	0.02	0.03	0.03	0.01	0.02	0.02	0.02	0.01	0.02	0.02	0.02
Max Mean Discrep ↓	0.03	0.03	0.08	0.08	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
Identifiability ↓	0.32	**0.25**	0.33	**0.26**	0.49	**0.37**	0.43	**0.34**	0.46	**0.33**	0.35	**0.30**
CLASSIFICATION METRICS												
Lin Syn ID ↑	0.69	**0.79**	0.67	**0.75**	0.90	0.91	0.87	0.89	0.88	0.88	0.79	**0.90**
Lin Syn OOD ↑	0.53	**0.79**	0.55	**0.75**	0.90	0.90	0.87	0.89	0.88	0.88	0.84	**0.90**
XGB Syn ID ↑	0.76	**0.83**	0.69	**0.73**	0.88	**0.91**	0.87	**0.90**	0.90	0.91	0.83	**0.93**
XGB Syn OOD ↑	0.60	**0.84**	0.60	**0.73**	0.88	**0.92**	0.87	**0.90**	0.90	0.90	0.86	**0.93**
REGRESSION METRICS												
XGB Syn ID ↑	-0.35	**0.23**	-0.32	**0.29**	0.33	**0.58**	0.23	**0.27**	0.12	**0.18**	0.12	**0.44**
XGB Syn OOD ↑	-0.72	**0.07**	-0.67	**0.14**	0.24	**0.53**	0.15	0.15	-0.09	**0.05**	0.12	**0.32**
Lin Syn MAPE ↓	0.43	0.40	0.90	**0.54**	0.36	**0.30**	0.36	0.35	0.35	0.36	0.34	**0.30**
XGB Syn MAPE ↓	0.47	**0.38**	0.53	**0.36**	0.32	**0.27**	0.37	**0.33**	0.36	0.36	0.36	**0.29**

of uncurated synthetic data beyond a certain point leads to accuracy degradation in classification tasks.

The curation process effectively reduces the negative impact on performance. The models trained with curated synthetic data achieve better ROC AUC scores at most proportions, often reaching or equaling the results obtained through real-only training. The TabPFN model's resistance to changes demonstrates its robustness and the additional advantages that come from curating synthetic data.

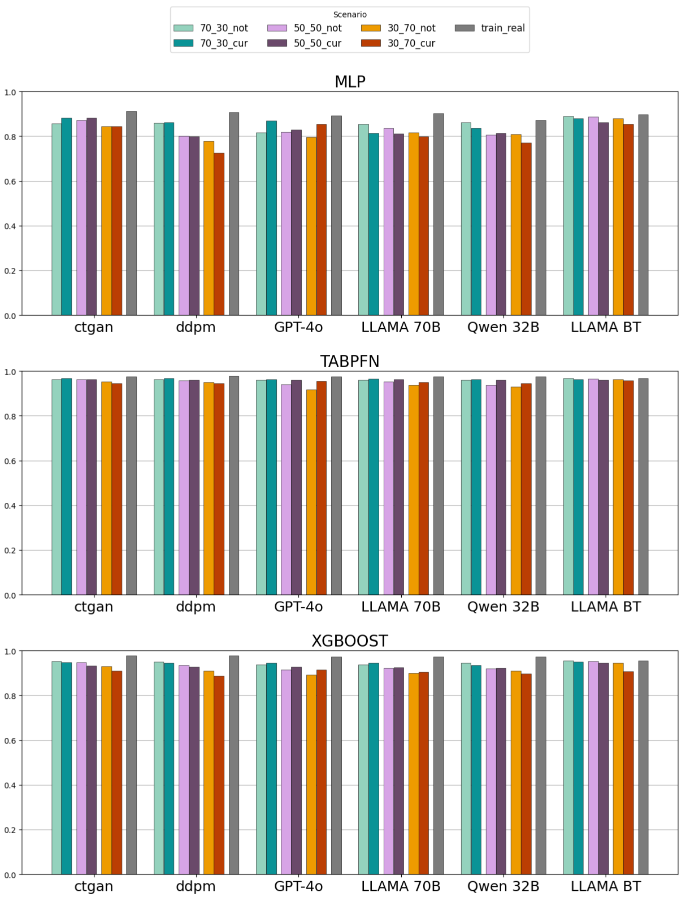

Figure 2. Results of predictive models trained on classification datasets. The performance is evaluated using ROC AUC

2) Regression Task Performance: Figure 3 illustrates a comparative analysis of regression models (MLP, TabPFN, XGBoost) by the coefficient of determination R^2 under similar conditions of data augmentation with synthetics. Scenario labels and groupings are similar to classification analysis. R^2 values with negative values are signed directly on the graph.

Regression results reveal increased variability and generally lower R^2 values under synthetic augmentation, with some scenarios demonstrating negative R^2 scores—indicative of model misfit—especially at higher synthetic proportions without curation.

Increasing the synthetic data ratio negatively impacts MLP and XGBoost regression models, with pronounced declines in R^2, confirming the susceptibility of regression tasks to synthetic data quality. However, the curation process markedly improves regression outcomes, reducing negative R^2 incidences and enhancing model fit, with notable gains observed for DDPM, GPT4o, and QWEN synthetic generators.

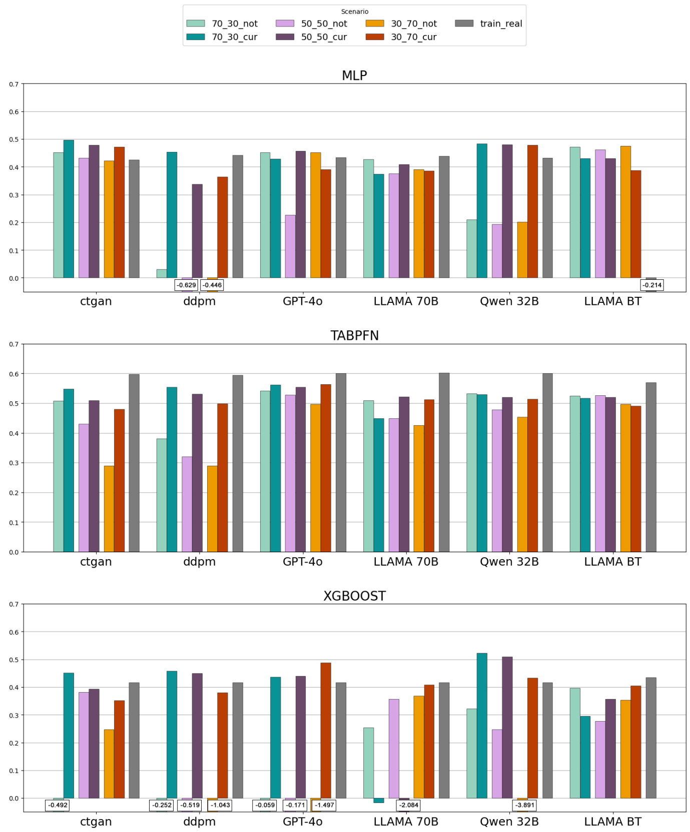

Figure 3. Results of predictive models trained on regression datasets. The performance is evaluated using R^2

TabPFN models showcase comparatively superior stability in regression metrics, reflecting inherent architectural advantages in handling tabular data and robustness to synthetic data noise.

Despite improvements post-curation, regression models trained on augmented datasets generally do not reach the baseline real-only performance, emphasizing the complexity of accurately modeling continuous outcomes using synthetic augmentation.

3) Joint Effects and Practical Considerations: Optimal augmentation involves balanced incorporation of curated synthetic data at moderate ratios (up to 50%), maximizing predictive performance while minimizing noise. High synthetic proportions without curation significantly compromise both classification and regression performance.

The results highlight the critical role of synthetic data quality control and model selection in leveraging augmented tabular datasets effectively. TabPFN emerges as a resilient architecture with minimal performance loss across conditions.

D. LLM Training on Synthetic Data

To enhance comprehension regarding the utilization of synthetic data in LLM training, the Llama-3B model was fine-tuned using various datasets comprising different proportions

Table V
EVALUATION OF SYNTHETIC DATA GENERATED BY FINE-TUNED LLAMA-3B AUGMENTED BY SYNTHETIC DATA BEFORE AND AFTER CURATION. **GREEN BOLD** METRICS INDICATE OVERALL IMPROVEMENT VIA SYNTHETIC AUGMENTATION, **RED BOLD** METRICS INDICATE DOWNGRADE.

Evaluation Metric	**100 real**	**70 real/30 syn**		**50 real/50 syn**		**30 real/70 syn**	
	no cur	no cur	cur	no cur	cur	no cur	cur
Close Values Prob ↑	0.68	0.34	0.34	0.34	0.33	0.34	0.34
Inv KL Divergence ↑	0.66	0.66	0.69	0.66	0.66	0.65	0.61
KS test ↑	0.85	0.85	0.85	0.84	0.84	0.84	0.83
JS Divergence ↓	0.02	0.02	0.02	0.02	0.02	0.02	0.02
Max Mean Discrep ↓	0.03	0.03	0.03	0.03	0.03	0.03	0.03
Identifiability ↓	0.35	0.03	0.03	0.03	0.02	0.02	0.02
CLASSIFICATION METRICS							
MLP Syn ID ↑	0.48	0.46	0.49	0.47	0.50	0.53	0.44
MLP Syn OOD ↑	0.52	0.45	0.51	0.49	0.51	0.56	0.45
XGB Syn ID ↑	0.83	0.55	0.54	0.51	0.54	0.52	0.49
XGB Syn OOD ↑	0.86	0.53	0.54	0.51	0.52	0.53	0.48
REGRESSION METRICS							
MLP Syn ID ↑	-1.47	-12.90	-25.58	-2.72	-10.49	-11.94	-4.99
MLP Syn OOD ↑	-1.73	-11.18	-56.19	-4.49	-22.07	-26.38	-5.30
XGB Syn ID ↑	0.12	-0.77	-0.66	-10.15	-0.85	-0.20	-0.59
XGB Syn OOD ↑	0.12	-1.27	-0.73	-12.83	-1.18	-1.24	-0.46

of augmented synthetic data. As shown in Table V, each column corresponds to a distinct mixture of real and synthetic data within the training dataset. The synthetic data employed in these mixtures was generated by sampling outputs from a model that had been exclusively trained on real data.

Based on the evaluation results, the augmentation of fine-tuned Llama-3B with synthetic data demonstrates a complex trade-off between privacy preservation and performance metrics, with curation showing mixed effects. The privacy metric, Identifiability, shows substantial improvement across all synthetic data mixtures, indicating enhanced privacy protection compared to using purely real data.

Statistical quality metrics including Inverse KL Divergence, KS test, JS Divergence, and Maximum Mean Discrepancy remain largely unchanged from original quality levels and appear unaffected by the proportion of synthetic data in the mixture, maintaining consistent values. The probability of close values between the real and synthetic data (Close Values Prob) shows decrease by 50% between all mixtures. The curation of synthetic data for training had minimal impact on these statistical measures, maintaining their stability.

The performance metrics exhibit significant degradation from original quality levels, particularly in regression tasks where MLP models show dramatically negative values (up to -56.19) compared to the baseline, and classification tasks where XGB models show decline by more than 30%. The effect of curation appears inconsistent—while curation slightly improved some results (e.g., Regression XGB Syn OOD improved from -12.83 to -1.18 in one mixture), it worsened others (e.g., Classification MLP Syn ID declined from 0.53 to 0.44). These results suggest that while training LLMs with synthetic data effectively enhances privacy preservation and maintains statistical fidelity, the process may introduce instability in model performance, potentially limiting the practical utility of synthetic data augmentation and curation for downstream tasks despite its privacy benefits.

VI. DISCUSSION AND LIMITATIONS

A. Implications of Data Curation for Scientific Data Creation

LLM privacy vulnerabilities appear intrinsic to LLM architectures rather than generation artifacts, as evidenced by consistent patterns across different model sizes and training approaches. The superior statistical modeling capabilities that enable LLMs to capture complex data distributions simultaneously increase their tendency to memorize training examples, creating an inherent tension between generation fidelity and privacy preservation.

Our results establish intelligent curation as a critical component for bridging the gap between synthetic data generation and practical AI utility. The systematic quality improvements achieved through automated curation processes address fundamental challenges in scientific data scarcity while maintaining computational efficiency. Moreover, the application of data curation enables the effective utilization of previously generated synthetic data, thereby enhancing overall quality metrics.

The demonstrated effectiveness of curated synthetic data for training both traditional machine learning models and large language models suggests broad applicability across scientific computing applications. This has particular relevance for data-limited scientific domains where experimental data collection faces significant constraints.

B. Limitation of Predictive ML models Training on Synthetic Data

The findings demonstrate fundamental limitations of training predictive models using synthetic data augmentation. Whereas classification models tend to preserve high levels of performance across different real-to-synthetic data ratios—particularly after curation—regression models are more vulnerable to quality loss. Negative R^2 values and performance degradation at high synthetic proportions highlight the difficulty in modeling continuous outcomes with imperfect synthetic data.

Curation of synthetic data largely alleviates harmful impacts, enhancing data fidelity and model robustness. However, performance is seldom comparable to that of models exclusively trained on real data, especially for regression tasks. In addition, overdependence on synthetic data without proper curation worsens model degradation.

Model robustness is not uniform: TabPFN has greater robustness to synthesized noise than MLP and XGBoost, implying architectural susceptibility to input data quality. Taken together, these results highlight the importance of diligent synthetic data quality management, suitable augmentation ratios, and task-specific model choice to maximize training results.

C. Limitation of LLM Training on Synthetic Data

Despite promising results, the fine-tuned LLM exhibits unstable performance metrics even with a relatively small augmentation of synthetic data. Under such circumstances, data curation is expected to be ineffective. Identifying the causes of fine-tuning instability may present an opportunity to develop a specialized curation algorithm tailored to this challenge.

VII. Conclusion

This work presents a two-stage approach combining LLM-based generation with curation that addresses critical data scarcity in scientific domains. Given the inherent privacy vulnerabilities of LLMs despite their superior generation capabilities, our curation pipeline becomes particularly important. The systematic quality assessment and refinement processes not only enhance data utility but also provide mechanisms to mitigate some privacy risks while preserving the statistical advantages of large-scale models.

Our systematic evaluation reveals four key findings that advance synthetic data generation for practical applications:

- **Large-scale LLMs consistently outperform specialized generative models** across statistical fidelity and downstream task utility. GPT-4o, LLaMA-70B, and Qwen-32B achieve superior PRDC scores (1.0 precision/recall vs. 0.4-0.8 for GANs/VAEs) and better predictive performance without requiring task-specific fine-tuning. However, this comes at a significant privacy cost, with LLMs showing 0.5 identifiability scores compared to 0.1-0.3 for traditional models.
- **Suggested curation pipeline is able to improve privacy** without degrading statistical distributions, effectively balancing privacy-utility trade-offs. Our curation pipeline delivers systematic improvements: 20% average reduction in privacy risks, up to 0.26 point gains in classification ROC AUC, and conversion of negative R^2 regression values to positive performance.
- **Synthetic data augmentation for predictive modeling results in expected quality degradation for most models, but reveals promising exceptions.** While traditional augmentation approaches show performance decline with increasing synthetic ratios, fine-tuned LLaMA 3.2-3B demonstrates superior and most stable results for classification tasks, enabling 50% data augmentation with minimal quality loss. For regression modeling, curation consistently improves quality metrics across most scenarios, effectively mitigating the negative impacts of synthetic data inclusion. TabPFN emerges as the most robust architecture to synthetic training data, maintaining stable performance regardless of synthetic data proportions and demonstrating superior resistance to synthetic data artifacts compared to MLP and XGBoost models.
- **LLM fine-tuning on synthetic data shows emerging privacy benefits while performance still reveals significant challenges.** While performance metrics decline substantially, identifiability scores drop dramatically from 0.35 to 0.02-0.03 with synthetic data inclusion, representing a 90% reduction in privacy risks. This suggests that synthetic data may be valuable for LLM training primarily in privacy-sensitive applications where performance trade-offs are acceptable, rather than as a general augmentation strategy for improving model capabilities.

All code and generation results are available in anonymous repository [1].

[1]https://anonymous.4open.science/r/Curation-B430/

References

[1] V. Borisov, K. Seßler, T. Leemann, M. Pawelczyk, and G. Kasneci, "Language models are realistic tabular data generators," *arXiv preprint arXiv:2210.06280*, 2022.

[2] S. Xu, C.-T. Lee, M. Sharma, R. B. Yousuf, N. Muralidhar, and N. Ramakrishnan, "Why llms are bad at synthetic table generation (and what to do about it)," *arXiv preprint arXiv:2406.14541*, 2024.

[3] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional gan," *Advances in neural information processing systems*, vol. 32, 2019.

[4] J. Jordon, J. Yoon, and M. Van Der Schaar, "Pate-gan: Generating synthetic data with differential privacy guarantees," 2018.

[5] J. Yoon, L. N. Drumright, and M. Van Der Schaar, "Anonymization through data synthesis using generative adversarial networks (ads-gan)," *IEEE journal of biomedical and health informatics*, vol. 24, no. 8, pp. 2378–2388, 2020.

[6] Y. Zhang, N. A. Zaidi, J. Zhou, and G. Li, "Ganblr: a tabular data generation model," in *2021 IEEE International Conference on Data Mining (ICDM)*. IEEE, 2021, pp. 181–190.

[7] Y. Abdalla, M. Taub, E. Hilton, P. Akkaraju, A. Milanovic, M. Orlu, A. W. Basit, M. T. Cook, T. Chakraborti, and D. Shorthouse, "Vect-gan: A variationally encoded generative model for overcoming data scarcity in pharmaceutical science," *arXiv preprint arXiv:2501.08995*, 2025.

[8] H. Ishfaq, A. Hoogi, and D. Rubin, "Tvae: Triplet-based variational autoencoder using metric learning," *arXiv preprint arXiv:1802.04403*, 2018.

[9] H. Akrami, S. Aydore, R. M. Leahy, and A. A. Joshi, "Robust variational autoencoder for tabular data with beta divergence," *arXiv preprint arXiv:2006.08204*, 2020.

[10] A. Kotelnikov, D. Baranchuk, I. Rubachev, and A. Babenko, "Tabddpm: Modelling tabular data with diffusion models," in *International conference on machine learning*. PMLR, 2023, pp. 17 564–17 579.

[11] A. A. Barr, R. Rozman, and E. Guo, "Generative adversarial networks vs large language models: a comparative study on synthetic tabular data generation," *arXiv preprint arXiv:2502.14523*, 2025.

[12] T. Liu, Z. Qian, J. Berrevoets, and M. van der Schaar, "Goggle: Generative modelling for tabular data by learning relational structure," in *The Eleventh International Conference on Learning Representations*, 2023.

[13] N. Seedat, N. Huynh, B. van Breugel, and M. van der Schaar, "Curated llm: Synergy of llms and data curation for tabular augmentation in ultra low-data regimes," 2023.

[14] L. Chen, S. Li, J. Yan, H. Wang, K. Gunaratna, V. Yadav, Z. Tang, V. Srinivasan, T. Zhou, H. Huang *et al.*, "Alpagasus: Training a better alpaca with fewer data," *arXiv preprint arXiv:2307.08701*, 2023.

[15] J. Saad-Falcon, O. Khattab, K. Santhanam, R. Florian, M. Franz, S. Roukos, A. Sil, M. A. Sultan, and C. Potts, "Udapdr: Unsupervised domain adaptation via llm prompting and distillation of rerankers," *arXiv preprint arXiv:2303.00807*, 2023.

[16] Q. Liu, M. Khalil, J. Jovanovic, and R. Shakya, "Scaling while privacy preserving: A comprehensive synthetic tabular data generation and evaluation in learning analytics," in *Proceedings of the 14th Learning Analytics and Knowledge Conference*, 2024, pp. 620–631.

[17] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Körfer, S. B. Hoo, R. T. Schirrmeister, and F. Hutter, "Accurate predictions on small data with a tabular foundation model," *Nature*, vol. 637, no. 8045, pp. 319–326, 2025.

[18] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen *et al.*, "Lora: Low-rank adaptation of large language models." *ICLR*, vol. 1, no. 2, p. 3, 2022.

[19] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar *et al.*, "Llama: Open and efficient foundation language models," *arXiv preprint arXiv:2302.13971*, 2023.

[20] Z. Qian, B.-C. Cebere, and M. van der Schaar, "Synthcity: facilitating innovative use cases of synthetic data in different data modalities," *arXiv preprint arXiv:2301.07573*, 2023.